CIVICTRACK

Smart Civic Issue Reporting Platform

Initial Capstone Project Report

Project Type	Capstone Project
Domain	Smart Governance / Civic Technology / AI / Web Application
Team Size	6 Members
Target Users	Citizens, Municipal Authorities, Panchayat Authorities, Government Officers and Admini trators

1. Project Overview

CivicTrack is a smart citizen-to-government platform designed to simplify the reporting, management, tracking, and resolution of local civic infrastructure problems.

Citizens frequently face problems such as potholes, damaged roads, broken streetlights, overflowing garbage bins, drainage problems, water leakage, damaged public property, and other infrastructure-related issues. CivicTrack creates a single digital platform where citizens can report these problems, provide photographs and location information, track complaint progress, view nearby reported issues, and support important problems through community upvoting.

Government authorities can use a centralized dashboard to view, validate, prioritize, assign, and monitor reported issues. An **AI-powered chatbot assistant** will also be integrated to guide citizens through reporting, explain statuses, help track complaints, and assist with platform navigation.

2. Problem Statement

Local civic infrastructure problems are common across India, including potholes, damaged roads, broken streetlights, overflowing garbage bins, drainage problems, water leakage, damaged public property, open manholes, broken footpaths, and illegal dumping.

Citizens may not know where to report an issue, which department is responsible, how to submit a proper complaint, whether the complaint was received, or when it will be resolved. Authorities also face duplicate complaints, difficulty identifying critical issues, assigning work, and monitoring resolution.

There is therefore a need for a **centralized, transparent, location-aware and intelligent civic issue management platform**. CivicTrack addresses this gap by connecting citizens and authorities through one digital system.

3. Proposed Solution

Provide a complete digital lifecycle for civic complaints.

Allow citizens to register, report issues, upload photographs/videos, provide locations, view nearby issues, upvote problems, track complaints, receive notifications, and use an AI assistant.

Allow administrators to review complaints, validate them, identify duplicates, detect spam, prioritize issues, assign departments/officers, update status, upload resolution proof, and monitor analytics.

4. Main Objectives

Citizen-centric reporting of civic infrastructure problems.

Centralized complaint management.

Location-based reporting and mapping.

Community participation through upvoting.

Intelligent prioritization using severity and community impact.

Transparent complaint tracking.

Administrative management and resolution verification.

AI assistance for easier platform interaction.

Data-driven civic planning and analytics.

5. Target Users

Citizens: report issues, upload media, track complaints, upvote issues, receive notifications, and use the AI assistant.

Government Administrators: review, validate, prioritize, assign, monitor, verify, and analyze complaints.

Department Officers: view assigned issues, update progress, schedule work, upload resolution evidence, and mark issues resolved.

6. Complete CivicTrack Workflow

```
CITIZEN
↓
Register / Login
↓
Report Civic Issue
↓
Description + Photo/Video
↓
GPS Location Tagging
↓
Issue Validation
■■■ Valid → Continue
■■■ Duplicate → Merge / Link
■■■ Spam / Invalid → Reject
↓
Live Map
↓
Community Upvoting
↓
Priority Calculation
↓
Admin Verification
↓
Department / Officer Assignment
↓
Pending → In Progress → Scheduled → Resolved
↓
Resolution Verification
↓
Citizen Notification
↓
Public Analytics
↓
Future Civic Planning
```

7. Detailed Issue Lifecycle

Step 1 – User Registration/Login: The citizen creates an account using the required credentials and accesses the CivicTrack dashboard after authentication.

Step 2 – Report an Issue: The citizen selects Report Issue and provides category, description, photograph/video, location, and optional information.

Step 3 – Location Tagging: The system associates the complaint with GPS coordinates and relevant ward, city/town, or panchayat information.

Step 4 – Issue Validation: The system checks required information, location validity, authenticity, duplicate reports, and possible spam. Reports are classified as valid, duplicate, or spam/invalid.

Step 5 – Duplicate Issue Detection: Similar reports from multiple citizens can be linked or merged into one primary issue, reducing duplicate work and improving community-impact measurement.

Step 6 – Live Civic Issue Map: An interactive map displays reported issues by category and allows users to view descriptions, status, location, upvotes, date reported, and progress.

Step 7 – Community Upvoting: Citizens can upvote existing issues. Higher community support can contribute to issue priority.

Step 8 – Priority Calculation: Priority can consider severity, upvotes, affected citizens, location importance, complaint age, safety risk, and issue category.

Step 9 – Administrative Verification: An administrator reviews the complaint, checks evidence and location, handles duplicates, sets priority, and assigns the appropriate department/officer.

Step 10 – Department and Officer Assignment: Issues are assigned to departments such as Road & Infrastructure, Electrical, Sanitation, Water Supply, Drainage, Public Works, or Waste Management.

Step 11 – Issue Status Lifecycle: Complaints progress through Pending, In Progress, Scheduled, Resolved, and Resolution Verification.

Step 12 – Resolution Verification: Authorities upload proof such as before/after photographs, completion details, date, officer information, and remarks.

Step 13 – Citizen Notification: Citizens receive updates when complaint status changes or the issue is resolved.

8. AI Assistant Chatbot

Purpose: The AI-powered Civic Assistant makes CivicTrack easier to use, especially for users who are unfamiliar with digital complaint systems.

Responsibilities: Guide users through reporting, suggest issue categories, explain complaint statuses, help track complaints, explain upvoting, answer common questions, explain department responsibilities, and assist with platform navigation.

Example: User: “I have a big pothole near my house. How can I report it?” The assistant can guide the user to Report Issue and request a photo, description, and location.

9. AI Chatbot Architecture

```
Citizen
↓
AI Chatbot Interface
↓
Natural Language Input
↓
AI / NLP Processing
■■■ FAQ
■■■ Complaint Tracking
■■■ Navigation
↓
CivicTrack API
↓
Database
↓
Response
↓
Citizen
```

10. Database Management

Core data includes Users, Issues, Issue Categories, Locations, Media References, Upvotes, Departments, Officers, Assignments, Complaint Status, Notifications, and Resolution information.

Entity	Representative Fields
Users	user_id, name, email, authentication reference, role, location

Entity	Representative Fields
Issues	issue_id, user_id, category, description, latitude, longitude, status, priority, created_at
Upvotes	upvote_id, issue_id, user_id, created_at
Departments	department_id, department_name, description
Assignments	assignment_id, issue_id, department_id, officer_id, assigned_at
Resolution	resolution_id, issue_id, proof, remarks, resolved_at
Notifications	notification_id, user_id, issue_id, message, created_at

11. Backend

Provides business logic and APIs for authentication, issue creation/retrieval/updating, validation, assignment, priority management, upvotes, status management, notifications, AI chatbot integration, and database communication.

12. Frontend

Citizen Interface: Landing Page, Login/Register, Citizen Dashboard, Report Issue, My Complaints, Complaint Details, Nearby Issues, Civic Map, Notifications, AI Assistant, Profile.

Admin Interface: Admin Dashboard, Issue Management, Issue Details, Map View, Department/Officer Assignment, Priority Management, Resolution Verification, and Analytics.

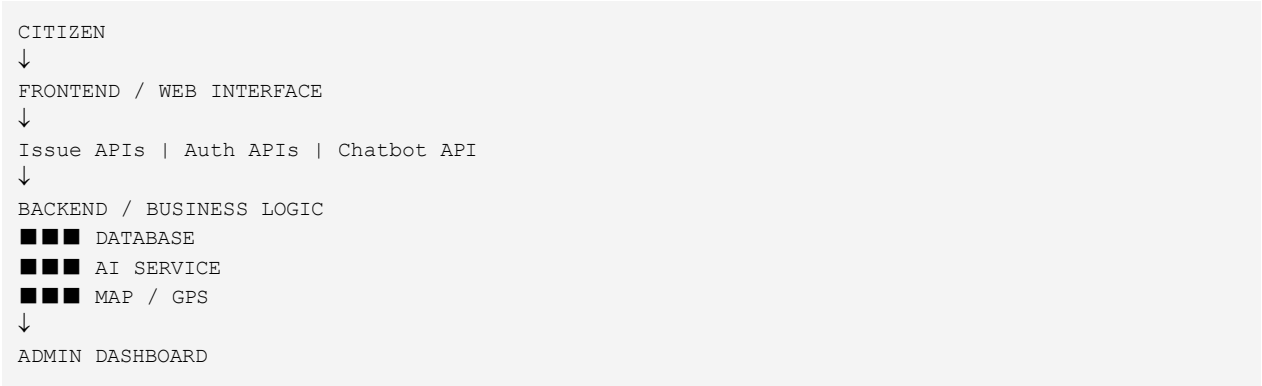
13. Admin Dashboard

Centralized metrics can include Total Issues, Open Issues, In Progress, Resolved, Critical Issues, Average Resolution Time, Issues by Category, Ward, Department, Resolution Rate, Pending Complaints, High-Priority Complaints, Most Reported Locations, and Community Participation.

14. Public Analytics

Selected public statistics can include total reported/resolved/pending/in-progress issues, category distribution, average resolution time, department performance, ward performance, and monthly resolution rate.

15. Proposed System Architecture



16. Team Structure and Responsibilities

Member	Primary Responsibility
Sairam	AI Chatbot & Database
Anurag	AI Chatbot & Database

Member	Primary Responsibility
Shubham	Backend Development
Harsh – Backend	Backend Development
Harsh – Frontend	Frontend Design & Development
Adarsh	Frontend Design & Development

Sairam – AI Chatbot & Database: AI chatbot development, chatbot integration, conversation flow, database design/schema, query optimization, and integration between chatbot and civic issue data.

Anurag – AI Chatbot & Database: AI assistant functionality, prompt/conversation design, FAQ/civic assistance logic, chatbot API integration, database implementation, data validation, and chatbot testing.

Subham – Backend Development: Backend APIs, authentication, issue management, complaint lifecycle, user management, upvotes, notifications, and backend testing.

Harsh – Backend Development: Admin APIs, department management, officer assignment, issue validation, priority management, resolution workflow, and frontend API integration.

Harsh – Frontend Design & Development: Citizen UI, report issue interface, complaint tracking, map interface, responsive design, and frontend-backend integration.

Adarsh – Frontend Design & Development: Admin dashboard, analytics interface, authentication screens, issue management screens, UI/UX improvements, and frontend testing.

17. Major System Modules

- Authentication Module
- Citizen Module
- Issue Reporting Module
- Location & Map Module
- Validation Module
- Duplicate Detection Module
- Community Upvoting Module
- Priority Management Module
- Admin Module
- Department & Officer Module
- Complaint Tracking Module
- Resolution Verification Module
- Notification Module
- AI Chatbot Module
- Analytics Module

18. Key Features

Citizen Features

- User registration/login
- Civic issue reporting

- Image/video upload
- GPS/location tagging
- Issue categorization
- Complaint tracking
- Nearby issue map
- Community upvoting
- Notifications
- AI assistant

Authority Features

- Admin dashboard
- Complaint validation
- Duplicate detection
- Spam detection
- Priority calculation
- Department assignment
- Officer assignment
- Status updates
- Resolution proof
- Performance monitoring
- Analytics

Smart Features

- AI chatbot
- Location-based issue discovery
- Community-based prioritization
- Potential AI-based duplicate detection
- Data-driven civic planning

19. Security Considerations

- Secure authentication
- Password hashing where applicable
- Role-based access control
- Input validation
- API authentication and authorization
- Protection against unauthorized complaint modification
- Secure file/media handling
- Database security

- Protection of citizen information

20. Expected Benefits

For Citizens

- Easy complaint reporting
- Transparent complaint tracking
- Faster communication
- Community participation
- Better visibility of local problems
- Simple AI-based assistance

For Authorities

- Centralized complaint management
- Better prioritization
- Reduced duplicate complaints
- Location-based issue monitoring
- Improved department coordination
- Performance analytics
- Data-driven decision making

For Communities

- Greater civic participation
- Better awareness of local issues
- Increased transparency
- Improved accountability

21. Innovation of the Project

CivicTrack combines several civic management features into one platform rather than functioning only as a complaint submission form.

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REPORT → VALIDATE → MAP → UPVOTE → PRIORITIZE → ASSIGN
→ TRACK → RESOLVE → VERIFY → ANALYZE
```

The AI civic assistant further simplifies interaction between citizens and the platform.

22. Future Scope

- AI-based duplicate detection
- Automatic issue classification from photographs
- Image-based pothole detection
- Severity estimation from images
- Spam/fake complaint detection
- Predictive civic issue analysis

- Integration with smart-city systems, IoT sensors, traffic systems, waste management, public lighting, and water monitoring
- Multilingual AI assistant supporting English, Hindi, Odia, Telugu, Bengali, Tamil and other regional languages
- Dedicated Android/iOS mobile application with GPS, camera integration, push notifications, offline complaint preparation, and mobile chatbot

23. Expected Project Outcome

The final prototype is expected to demonstrate the complete civic issue lifecycle from reporting to resolution and analytics.

```
Citizen:
Register → Report → Upload Photo → Add Location → Submit
→ View on Map → Track Status → Receive Resolution

Authority:
Receive → Validate → Check Duplicate → Set Priority
→ Assign Department → Assign Officer → Monitor Progress
→ Verify Resolution → Generate Analytics
```

24. Conclusion

CivicTrack is proposed as a smart, transparent and citizen-centric civic issue management platform that bridges the communication gap between citizens and local government authorities. The platform provides a complete workflow for reporting, validating, prioritizing, assigning, tracking and resolving civic infrastructure problems.

The integration of location services, community upvoting, administrative dashboards, analytics and an AI-powered civic assistant makes CivicTrack more than a traditional complaint management system. The project can provide a strong foundation for future expansion into smart-city applications, multilingual civic assistance, AI-based issue detection, predictive analytics and integration with government infrastructure systems.

Ultimate Goal: Make civic issue reporting simpler for citizens, more manageable for authorities, and more transparent for communities.